MINISTRY OF EDUCATION AND RESEARCH OF THE REPUBLIC OF MOLDOVA

TECHNICAL UNIVERSITY OF MOLDOVA

Faculty of Computers, Informatics and Microelectronics
Department of Software Engineering and Automation

LABORATORY WORK No. 2

Cryptanalysis of Monoalphabetic Ciphers

Variant V11

Performed by:
Student, group FAF-243
Andrei Elciscev

Checked by:
Zaica Maia

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1 Objective

The objective of this laboratory work is to study the cryptanalysis of monoalphabetic ciphers using letter frequency analysis and to apply this method to decrypt an intercepted message written in English.

The specific objectives are:

- to understand how monoalphabetic substitution ciphers work;
- to analyze the frequency of letters in an encrypted text;
- to identify possible correspondences between ciphertext and plaintext letters;
- to use word patterns and context to verify substitutions;
- to reconstruct the original message for Variant V11.

2 Theoretical Background

2.1 Letter Frequency Analysis

A monoalphabetic substitution cipher is an encryption system in which each plaintext letter is replaced with a corresponding ciphertext letter. In a simple monoalphabetic substitution, the same plaintext letter is always replaced by the same ciphertext letter.

The main weakness of this system is that it preserves the relative frequency pattern of letters. In a sufficiently long encrypted text, the frequency distribution of ciphertext letters reflects, to some extent, the distribution of letters in the language of the original message.

In English, the letters E, T, A, O, I, and N occur frequently. Therefore, if a particular letter appears very often in a ciphertext, it may represent one of these common plaintext letters.

However, this assumption is not always correct. Letter frequencies vary between texts depending on their subject, length, and vocabulary. For this reason, frequency analysis should be combined with the examination of word patterns, letter pairs, three-letter combinations, and the overall meaning of the message.

2.2 Frequency Analysis Methodology

A frequency analysis attack can be performed using the following steps:

1. Identify all letters occurring in the ciphertext.
2. Count the occurrences of each letter.
3. Sort the letters in descending order of frequency.
4. Compare the resulting distribution with typical English letter frequencies.

5. Form hypotheses about possible substitutions.

6. Apply the substitutions and examine the resulting words.

7. Verify the hypotheses using common words, grammatical structure, and the context of the message.

8. Repeat the process until the text can be reconstructed and the substitutions are sufficiently verified.

In English, words such as *the*, *and*, *was*, *under*, and *with* can help verify substitution hypotheses. Short words and repeated patterns can also provide useful clues.

2.3 Mean Frequency of Letters in the English Alphabet

Frequency analysis relies on the fact that letters do not occur equally often in English texts. The following table presents the approximate average frequency of each letter in English, expressed as a percentage of all letters in a typical text. These values serve as a reference when analyzing the ciphertext and identifying possible substitutions.

Table 1: Approximate Letter Frequencies in English

Letter	**Freq. (%)**	**Letter**	**Freq. (%)**	**Letter**	**Freq. (%)**	**Letter**	**Freq. (%)**
A	8.17	H	6.09	O	7.51	U	2.76
B	1.49	I	6.97	P	1.93	V	0.98
C	2.78	J	0.15	Q	0.10	W	2.36
D	4.25	K	0.77	R	5.99	X	0.15
E	12.70	L	4.03	S	6.33	Y	1.97
F	2.23	M	2.41	T	9.06	Z	0.07
G	2.02	N	6.75				

As shown in Table 1, the most frequent letters in English are typically E, T, A, O, I, and N. Conversely, letters such as Q, X, Z, and J occur relatively rarely. When analyzing a monoalphabetic substitution cipher, the most frequent ciphertext symbols can therefore be compared with these common English letters to suggest possible substitutions. However, frequency alone does not prove a substitution; letter patterns, common words, and the consistency of the resulting plaintext must also be considered.

3 Intercepted Ciphertext

Variant V11 consists of a message encrypted using a monoalphabetic substitution cipher. It is known that the original message was written in English and that only letters were encrypted. Spaces, punctuation marks, and other characters remain unchanged.

The ciphertext begins with the following sequence:

```
IVTSZNGW rtp dgovi pxvjv.
```

The message contains several paragraphs describing historical events in seventeenth-century France, including the siege of Réalmont and the activities of the cryptanalyst Antoine Rossignol.

The length of the message provides sufficient material for frequency analysis, while the preserved spaces allow the structure of individual words to be examined.

4 Cryptanalysis Process

4.1 Step 1: Examining the Beginning of the Ciphertext

The cryptanalysis begins with the first words of the intercepted message:

```
IVTSZNGW rtp dgovi pxvjv.
```

The spaces and punctuation have been preserved, so the lengths of the words provide useful information. The first word contains eight letters, followed by three-letter, five-letter, and five-letter words. Word length alone is not sufficient to determine the plaintext, but it helps narrow down possible interpretations.

The message is known to concern seventeenth-century France and the cryptanalyst Antoine Rossignol. This historical context can help verify a hypothesis, although the substitution must also remain consistent throughout the ciphertext.

A promising interpretation of the opening is:

REALMONT was under siege.

This gives us an initial hypothesis that can be tested against other parts of the intercepted message.

4.2 Step 2: Examining Common Digraphs and Trigraphs

Before assigning substitutions to individual letters, it is useful to examine repeated groups of two and three letters. In English, common digraphs include *th*, *he*, *in*, *er*, and *an*. Common trigraphs include *the*, *ing*, and *and*.

The ciphertext contains several recurring patterns. For example, the word `Wqv` appears frequently and has the same three-letter pattern each time. The word `wqtw` also occurs repeatedly.

If the hypothesis that `Wqv` represents *the* is correct, the following substitutions can be proposed:

Table 2: Initial hypotheses from repeated letter patterns

Ciphertext	Hypothesis	Reason
`Wqv`	*the*	A frequent three-letter word that fits the English text.
`wqtw`	*that*	A repeated four-letter pattern consistent with the word *that*.
`wqvf`	*they*	A four-letter word that becomes meaningful under the proposed substitutions.
`xw`	*it*	A short, frequently occurring word that can be checked in context.

These are hypotheses rather than independent proofs. They become more convincing when the same substitutions produce recognizable words in multiple places in the message.

4.3 Step 3: Establishing the First Substitutions

The opening phrase provides the first set of letter correspondences. Assuming that the first four words have been correctly identified, the mapping can be constructed letter by letter.

Table 3: First substitutions obtained from the opening phrase

Ciphertext Letter	Plaintext Letter	Example
I	R	`IVTSZNGW` → REALMONT
V	E	`IVTSZNGW` → REALMONT
T	A	`IVTSZNGW` → REALMONT
S	L	`IVTSZNGW` → REALMONT
Z	M	`IVTSZNGW` → REALMONT
N	O	`IVTSZNGW` → REALMONT
G	N	`IVTSZNGW` → REALMONT
W	T	`IVTSZNGW` → REALMONT
R	W	`rtp` → was
P	S	`rtp` → was
D	U	`dgovi` → under
O	D	`dgovi` → under
X	I	`pxvjv` → siege
J	G	`pxvjv` → siege

Applying these substitutions to the opening gives:

`IVTSZNGW rtp dgovi pxvjv.`
⇓
REALMONT was under siege.

The first word is now recognizable, and the remaining words form a grammatically plausible sentence. This provides a starting point for extending the substitution mapping.

4.4 Step 4: Applying the First Substitutions to the Next Words

The next portion of the ciphertext is:

```
Wqv infts tizf, dgovi Qvgif XX nc Andiang, Uixghv nc Hngov, qto
```

The substitutions already identified can be applied to the text. The result is partially readable, but some letters still require additional analysis. In particular, the repeated word `Wqv` supports the hypothesis that it represents *the*.

The phrase can be interpreted as:

The royal army, under Henry II of Bourbon, Prince of Condé, had

This interpretation supplies additional correspondences, including `Q` → `H`, `F` → `Y`, and `X` → `I`. It also provides further opportunities to test the existing mapping.

Table 4: Additional substitutions inferred from the next phrase

Ciphertext Letter	Plaintext Letter	Example
Q	H	`Wqv` → the; `Qvgif` → Henry
F	Y	`infts` → royal
X	I	`XX` → II
A	B	`Andiang` → Bourbon
U	P	`Uixghv` → Prince
H	C	`Hngov` → Condé
C	F	`nc` → of

The proposed mappings must be checked against the rest of the message. A substitution should never be changed merely to make one word look correct if it contradicts another word that has already been verified.

4.5 Step 5: Decrypting a Longer Fragment

After incorporating the additional substitutions, a longer fragment can be reconstructed. The following example illustrates the progression from ciphertext to readable plaintext:

Ciphertext:

```
Wqv infts tizf, dgovi Qvgif XX nc Andiang, Uixghv nc Hngov, qto
```

Decrypted text:

The royal army, under Henry II of Bourbon, Prince of Condé, had

This stage demonstrates why it is useful to analyze the message beyond its first sentence. Each additional recognizable word provides more evidence for or against the current substitution mapping.

The next ciphertext sequence is:

`xgkvpwvo xw tw otrg Rvogvpotf, Tuixs 19, 1628.`

The same mapping produces:

invested it at dawn Wednesday, April 19, 1628.

The date and historical context make this interpretation plausible. The word `xgkvpwvo`, for example, can be reconstructed as *invested*, providing additional evidence for several substitutions.

4.6 Step 6: Using Repeated Words to Confirm the Mapping

Once several words have been identified, the mapping can be tested against repeated words and phrases elsewhere in the ciphertext.

For example, the ciphertext word `wqvf` occurs in sentences where the plaintext uses *they*. The repeated sequence `wqtw` is consistent with *that*. The word `xw` appears in contexts where *it* is expected.

These examples are useful because they test the same letter correspondences in different contexts. If the mapping is correct, the same ciphertext letter must always produce the same plaintext letter, regardless of its position in a word.

The message also contains recognizable names and historical terms, including *Huguenots*, *Condé*, *Richelieu*, *La Rochelle*, and *Antoine Rossignol*. These terms provide contextual confirmation once the letter substitutions have made them readable.

4.7 Step 7: Completing the Substitution Table

The mapping is extended by recording every confirmed ciphertext-to-plaintext correspondence. The following table summarizes the substitutions established from the opening phrases. It is an initial working table; the remaining letters should be added as they are verified throughout the full ciphertext.

Table 5: Working substitution table

Cipher	**Plain**	**Cipher**	**Plain**	**Cipher**	**Plain**
A	B	I	R	Q	H
C	F	J	G	R	W
D	U	N	O	S	L
F	Y	O	D	T	A
G	N	P	S	U	P
H	C	V	E	W	T
X	I	Z	M		

V	W	G	X	T	N	P	I	Q	O	S	D	H	J	C	Z	F	U	R	A	K	L	Y	B	M	E
434	325	256	251	243	233	221	193	192	149	132	110	104	75	72	72	68	65	53	44	28	10	6	3	3	2
13.0	9.7	7.7	7.5	7.3	7.0	6.6	5.8	5.7	4.5	3.9	3.3	3.1	2.2	2.2	2.2	2.0	1.9	1.6	1.3	0.8	0.3	0.2	0.1	0.1	0.1
E	T	N	I	A	O	S	R	H	D	L	U	C	G	F	M	Y	P	W	B	V	K	X	L	Z	J

Figure 1: Substitution table of the deciphered text

4.8 Step 8: Reconstructing and Verifying the Complete Message

After the substitution table has been sufficiently expanded, it can be applied to the entire ciphertext. The resulting text is checked for grammatical correctness, recognizable names, dates, and continuity of meaning.

The verification process includes the following checks:

1. Confirm that repeated ciphertext letters produce the same plaintext letters throughout the message.
2. Check that common digraphs and trigraphs form plausible English combinations.
3. Verify short words and repeated word patterns.
4. Compare the reconstructed words with the historical context.
5. Revisit any substitution that produces contradictions.
6. Read the complete message to identify remaining errors caused by transcription or incorrect hypotheses.

Frequency analysis is most effective when combined with word patterns and contextual reasoning. A high-frequency ciphertext letter may correspond to a common English letter, but frequency alone cannot establish the correct substitution. The strongest evidence comes from a mapping that consistently explains the entire intercepted message.

5 Decrypted Message

The following passage represents pretty formatted text of the deciphered message.

> REALMONT was under siege. The royal army, under Henry II of Bourbon, Prince of Condé, had invested it at dawn Wednesday, April 19, 1628. But the Huguenots, inside the battlements of the little town in southern France, were putting up a stiff defense. They cannonaded Condé from a tower and contemptuously rejected his demands that they surrender, saying that they would die instead. Condé brought up five big cannon from Albi, a dozen miles away, and on Sunday ranged them in an ominous line facing Realmont.
>
> That same day his soldiers captured an inhabitant of the town who was trying to carry an enciphered message to Huguenot forces outside. None of Condé's men could unriddle

it, but during the week the prince learned that it might be solved by the scion of a leading family of Albi who was known to have an interest in ciphers.

Condé sent him the cryptogram. The young man solved it on the spot. It revealed that the Huguenots desperately needed munitions and that, if they were not supplied, they would have to yield. This was news indeed, for despite the destruction of a number of houses by the Catholic batteries, the town was continuing to resist stoutly with no sign of surrender. Condé returned the cryptogram to the inhabitants, and on Sunday, April 30, 1628, though its fortifications were still unbreached and its defenses still apparently adequate for a long siege, Realmont suddenly and unexpectedly capitulated. With this dramatic success began the career of the man who was to become France's first full-time cryptologist: the great Antoine Rossignol.

When word of the incident reached Cardinal Richelieu, the astute and able Gray Eminence of France, he at once attached this useful talent to his suite. Rossignol proved his worth almost immediately. The Catholic armies under Richelieu surrounding the chief Huguenot bastion of La Rochelle intercepted some letters in cipher, which the young codebreaker of Albi read with ease. He told His Eminence that the starving citizens were eagerly awaiting help that the English had promised to send by sea. When the fleet arrived, the primed guardships and forts so intimidated it that it stood off the port's entrance and made no serious attempt to force a passage. A month later, the city capitulated in full sight of the English vessels—and the great French tradition of expertise in cryptology had been founded.

Rossignol very quickly established himself in the royal service. By 1630, his solutions had made him rich enough to build a small but elegant chateau at Juvisy, 12 miles south of Paris, later surrounding it with a charming informal garden designed by Le Notre, the gardener of Versailles. Here Louis XIII stopped to visit the young crypt-analyst in 1634, 1635 and 1636 on his returns to Paris from Fontainebleau.

In the swashbuckling court of that monarch, and then in the resplendent one of Louis XIV, Rossignol served with an extraordinary facility. The stronghold of Hesdin surrendered a week sooner than it otherwise would have because he solved an enciphered plea for help, and then composed a reply in the same cipher telling the townspeople how futile their hopes were. How many other towns he compelled to surrender, how many diplomatic coups he made possible, how many betrayals he uncovered among the great nobles in those days of shifting allegiances, he never discussed.

This reticence caused some at the court to charge that he never actually solved a single cipher, and that the cardinal spread inflated rumors about his abilities to discourage would-be conspirators. But in fact Richelieu was frequently telling his subordinates such things as, "It is necessary to make use, in my opinion, of the letters of the man who has been arrested by the civil authorities at Mezieres, that is to say, have them put into Rossignol's hands to see if there is something important in them." Or, eight years later, in 1642, writing to Messieurs de Noyers and de Chavigny: "I saw, in some extracts, that Rossignol sent me, a truce negotiation of the King of England with the Prince of Orange; I do not think that it can have any effect, but . . . it is up to you, gentlemen, to

keep your eyes peeled."

6 Conclusions

This laboratory work examined letter frequency analysis and its application to the cryptanalysis of a monoalphabetic substitution cipher.

The cryptanalysis process involves identifying letter frequencies, forming hypotheses about substitutions, and verifying them by examining word patterns and context. Frequency analysis provides useful clues, but it is not sufficient on its own to guarantee the correct decryption of every message.

The exercise demonstrates the vulnerability of monoalphabetic ciphers when sufficiently long messages are available. Since each letter is replaced consistently, statistical and linguistic characteristics of the original text remain visible in the ciphertext.

In conclusion, combining letter frequency analysis with word-pattern examination and contextual reasoning allows the original message to be progressively reconstructed. This illustrates why simple monoalphabetic substitution does not provide sufficient cryptographic security.